

Beyond Detection of Explainable AI for Pneumonia Diagnosis

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I. LITERATURE REVIEW

Deep learning has transformed chest X-ray (CXR)–based diagnosis of pneumonia and COVID–19, yet the opacity of these models remains a central barrier to clinical adoption. Recent work has increasingly focused on Explainable AI (XAI) to make predictions transparent, clinically meaningful, and trustworthy. This section reviews twelve key studies spanning: (i) large-scale CXR datasets, (ii) high-performing CNN and transformer architectures, (iii) weakly supervised localization, (iv) severity and trajectory modelling, and (v) human-in-the-loop XAI. These collectively motivate our choice of architectures, datasets, and XAI-ICP–based evaluation.

A. Dataset Foundations and Weakly Supervised Localization

Wang *et al.* introduced ChestX-ray8 (later extended to ChestX-ray14), a hospital-scale dataset of 108,948 frontal CXR images from 32,717 patients with eight thoracic disease labels automatically mined from radiology reports via NLP and negation detection [1]. They fine-tuned ImageNet-pretrained CNNs (AlexNet, VGG16, GoogLeNet, ResNet-50) in a unified framework with global pooling and multi-label loss, and showed that even when trained solely on image-level labels, these models learned class-specific activation maps that weakly localize disease. ResNet-50, for example, achieved AUCs of 0.81 for cardiomegaly and 0.79 for pneumothorax. A small subset of 1,600 bounding boxes provided ground truth for localization, evaluated by IoU and IoBB criteria. Although localization performance was modest by current XAI standards, this work established the feasibility of weakly supervised detection and localization on noisy, large-scale CXR data and underpins much of today’s thoracic XAI research.

Building on this, Shahi and Bagale investigated weakly supervised pneumonia localization using seven architectures—ResNet-18/50, DenseNet121, EfficientNet-B0, MobileNetV2/V3, and ViT-B/16—on the Kermany pediatric CXR dataset [2]. To avoid data leakage, they re-partitioned the dataset at the patient level (70/15/15 train/val/test). They used focal loss and random oversampling to mitigate heavy class imbalance and applied extensive augmentation. All models reached high test accuracies (96–98%) and ROC-AUCs ≥ 0.995 ; ResNet-18 and EfficientNet-B0 achieved 98% accuracy and F1 = 0.987, while MobileNet-V3 achieved 97% accuracy with the lowest parameter count, highlighting its suitability for mobile or edge deployment. Grad-CAM

was employed across CNNs, and adapted to patch tokens for ViT, to generate heatmaps. Qualitative inspection revealed appropriate focusing on opacified lung regions in pneumonia cases, and more diffuse activations in normals. To quantify anatomical plausibility, they proposed a Lung Attention Ratio (LAR)—the fraction of Grad-CAM activation power falling within a coarse lung ROI—showing that MobileNet-V3 and ViT-B/16 had the most lung-focused attention. This study demonstrates that modern, lightweight and transformer architectures can deliver both strong classification and clinically reasonable localization under weak supervision, and suggests simple in-lung attention metrics as an objective complement to visual assessment.

B. High-Performing CNN Architectures and Multi-Class Respiratory Tasks

Several works have evaluated different CNN families (residual, dense, efficient) and transfer learning strategies for pneumonia and COVID–19.

Islam *et al.* evaluated Xception, VGG19, and ResNet50 on a Kaggle CXR dataset (21,165 images) for both binary (COVID vs. normal) and four-class classification (COVID, normal, lung opacity, viral pneumonia) [3]. Using 512×512 input resolution and fine-tuning, Xception and VGG19 achieved 97.5% accuracy with AUCs of 1.0 and 0.99, respectively, in the binary task, outperforming ResNet50 (93.3%). In the four-class task, Xception performed best (93% accuracy), followed by VGG19 (92%), with ResNet50 trailing (75%). Layer-wise Grad-CAM showed that deeper blocks in Xception captured higher-order disease patterns, and that many VGG19 errors arose from attention to non-pathological regions, illustrating how XAI can reveal failure modes not apparent in scalar metrics. The authors also found that higher spatial resolution systematically improved performance, motivating our use of higher-resolution inputs where feasible.

Mahamud *et al.* proposed an enhanced DenseNet201 pipeline for multi-disease classification (COVID–19, pneumonia, tuberculosis, normal) from CXR [4]. A sophisticated preprocessing chain (denoising autoencoders, CLAHE, gamma correction) substantially improved image quality. Through ablation, they found that AveragePooling2D, batch size 64, and learning rate 10^{-6} minimized validation loss. The resulting DenseNet201 model achieved 99.2% accuracy with 99% precision and recall, outperforming EfficientNetB0 (94%),

InceptionV3 (92%), and LeNet (91%). To interpret decisions, they used SHAP, LIME, Grad-CAM, and Grad-CAM++; SHAP most effectively highlighted disease-specific features; LIME provided local superpixel-based explanations; Grad-CAM/Grad-CAM++ produced class-discriminative heatmaps, with Grad-CAM++ improving localization by leveraging both positive and negative gradients. Deploying the model in an Android app further illustrated the practicality of pairing high-performance DenseNet models with multi-XAI evaluation and mobile deployment.

Goyal and Singh took a hybrid approach combining classical computer vision with deep learning to distinguish COVID-19, bacterial pneumonia, viral pneumonia, and normal [5]. They applied median filtering and histogram equalization, then used a custom region-growing algorithm and morphology to extract lung ROIs. From these, they computed handcrafted features—HOG, GLCM texture, intensity statistics, and geometric moments—then applied feature normalization (min-max and a log-based “robust” scaling). SVM, KNN, ANN, ensemble methods, and an RNN-LSTM (F-RNN-LSTM) were compared on C19RD and CXIP datasets. Robust normalization combined with F-RNN-LSTM yielded the best performance (up to 95.0% and 94.3% accuracy), while reducing training and inference time relative to CNN baselines such as ResNeXt-50 and COVIDetectionNet. Although their focus was not modern XAI, the explicit ROI extraction and feature-level analysis provide structural interpretability.

C. Severity Assessment, Trajectory Modelling, and Latent-Space XAI

Beyond binary disease detection, several studies focus on modelling severity and temporal evolution using CXR.

Zandehshahvar *et al.* developed a CNN-based system to classify COVID-19 severity (normal, mild, moderate, severe) from 1,208 CXRs of 396 RT-PCR-positive patients from Emory Healthcare [6]. Using a VGG16 backbone pre-trained on the RSNA pneumonia dataset and fine-tuned on Emory data, they achieved average AUCs of 0.97, 0.90, 0.90, and 0.95 for the four severity classes across ten bootstrap test sets. Grad-CAM saliency maps showed strong overlap between model attention and radiologist-annotated regions. They also used APoZ-based pruning to visualize dominant filters and UMAP to embed the penultimate layer into a 2D latent space, where disease trajectories across longitudinal CXRs were plotted, distinguishing paths leading to recovery versus deterioration.

Sheu *et al.* proposed the XAI-ICP framework, designed for interpretable pneumonia infection classification with severity analysis [7]. DenseNet121 is pre-trained on NIH ChestX-ray and then fine-tuned on a local hospital dataset (TCVGH) plus VinDr-CXR. The model focuses on infiltrate, effusion, and cardiomegaly, and produces section-wise severity scores. SHAP is used to highlight top contributing features and regions, while radiologists label features and image quality. Independent learning on TCVGH achieved 92.14% accuracy and AUC 93.57%; adding transfer learning from VinDr raised performance to 93.29% accuracy and AUC 95.43%. XAI-ICP

incorporates human-in-the-loop feedback and an XAI scoring system (transparency, explainability, human-interaction, transferability), motivating our adoption of XAI-ICP as a unifying interpretability layer.

D. Comparative XAI Methods and Human-Centered Evaluation

Zou *et al.* developed an ensemble XAI strategy for mortality prediction in community-acquired pneumonia and COVID-19 patients, combining Grad-CAM++ and SHAP via kernel ridge regression and comparing it against five individual methods [8]. Using a visual explainability checklist (decision impact ratio, confidence impact ratio, localization, radiologist trust), they found the ensemble produced more stable and clinically trusted heatmaps, with a radiologist trust score of 70.2% over 76 true positives. LIME’s high quantitative absence-impact score did not correspond to clinically meaningful explanations, underscoring the need for qualitative, clinician-centered XAI evaluation.

Ennab and Mcheick compared Pixel Level Interpretability (PLI) against Grad-CAM on VGG19 models for pneumonia detection [9]. While Grad-CAM achieved higher accuracy and precision, PLI attained 100% recall and high clinician satisfaction for early-stage localization. Both showed moderate explanation stability ($SSIM \approx 0.439$). The authors recommend a hybrid strategy combining PLI’s fine-grained localization with Grad-CAM’s global overview.

Erukude *et al.* applied Grad-CAM to ResNet50 and DenseNet121 models on brain MRI and CXR pneumonia datasets [10]. DenseNet121 outperformed ResNet50 in accuracy and produced more anatomically coherent attention maps, while ResNet50 often activated non-pathological regions. Khaidos *et al.* compared DenseNet121 and EfficientNet-B0 on pediatric pneumonia detection, using both Grad-CAM and LIME [11]. EfficientNet-B0 achieved higher accuracy, F1, and MCC, with very high recall (> 0.99) for both models. Heatmaps consistently localized clinically relevant lung regions.

Ahmed *et al.* provided a systematic review of XAI in medical imaging, categorizing methods into saliency/heatmap-based, attention-based, model-agnostic, and rule-based/symbolic families [12]. They highlighted gaps such as the lack of standardized interpretability metrics, limited human-in-the-loop studies, nascent work on GNNs and multimodal transformers, and weak regulatory integration (FDA, EU AI Act). Their analysis supports our use of multiple XAI families and clinically grounded evaluation.

E. Synthesis

Taken together, these twelve studies show that: (i) large-scale, weakly labeled datasets like ChestX-ray14 can support robust multi-label classification and weakly supervised localization [1]; (ii) modern dense and efficient architectures (DenseNet, EfficientNet, MobileNet, ViT) often yield both higher accuracy and more clinically aligned explanations than older backbones [3], [4], [10], [2]; (iii) ensembles and hybrid

XAI strategies can outperform single methods in both quantitative and subjective evaluation [8], [9], [4]; (iv) severity modelling and latent-space visualization can provide rich, patient-specific trajectories rather than static binary labels [6], [7]; and (v) human-centered, regulatory-aware XAI frameworks such as XAI-ICP are essential for clinical translation [12], [7]. These insights motivate our adoption of a diverse 12-model evaluation suite unified under the XAI-ICP framework and applied to the NIH ChestX-ray14 dataset, with multi-method XAI evaluation and severity-aware, anatomy-aware assessment of explanations.

II. DATA SPECIFICATIONS

A. Dataset Overview

- **Name:** ChestX-ray14 (NIH Chest X-ray Dataset)
- **Source:** Kaggle (Community-hosted version)
- **Access URL:** <https://www.kaggle.com/datasets/nih-chest-xrays/data>

B. Dataset Details

- **Total Images:** 112,120 frontal-view chest X-rays
- **Unique Patients:** 30,805
- **Image Format:** PNG
- **Resolution:** 1024×1024 pixels
- **Label Type:** Multi-label classification
- **Disease Categories:** 14 thoracic conditions

III. ALGORITHMS AND MODELS

- **Baseline and Traditional Models:** K-Nearest Neighbors (KNN) is used to establish a performance baseline and assess feature separability, while a custom CNN serves as a simple deep-learning baseline without pre-trained weights.
- **Residual and Dense Convolutional Networks:** ResNet18, ResNet50, and ResNet152V2 investigate the impact of depth and skip connections on detecting subtle pathologies, whereas DenseNet-121 exploits dense connectivity and feature reuse for improved gradient flow and accuracy.
- **Efficient and Mobile Architectures:** MobileNetV2 and MobileNetV3 employ depthwise separable convolutions for lightweight, resource-efficient deployment. EfficientNet-B0 applies compound scaling to balance accuracy and computational cost in CXR classification.
- **Advanced Vision Transformers:** ViT-B16 models long-range global dependencies across the image, capturing contextual relationships that complement local convolutional features.
- **XAI-ICP Interpretability Framework:** The XAI-ICP (Explainable AI – Integrated Class Probability) framework is applied across all models to provide unified interpretability via class-discriminative heatmaps and attribution scores focused on clinically relevant lung regions.

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